

Metrics

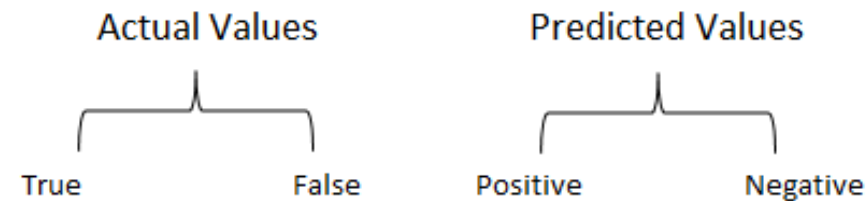
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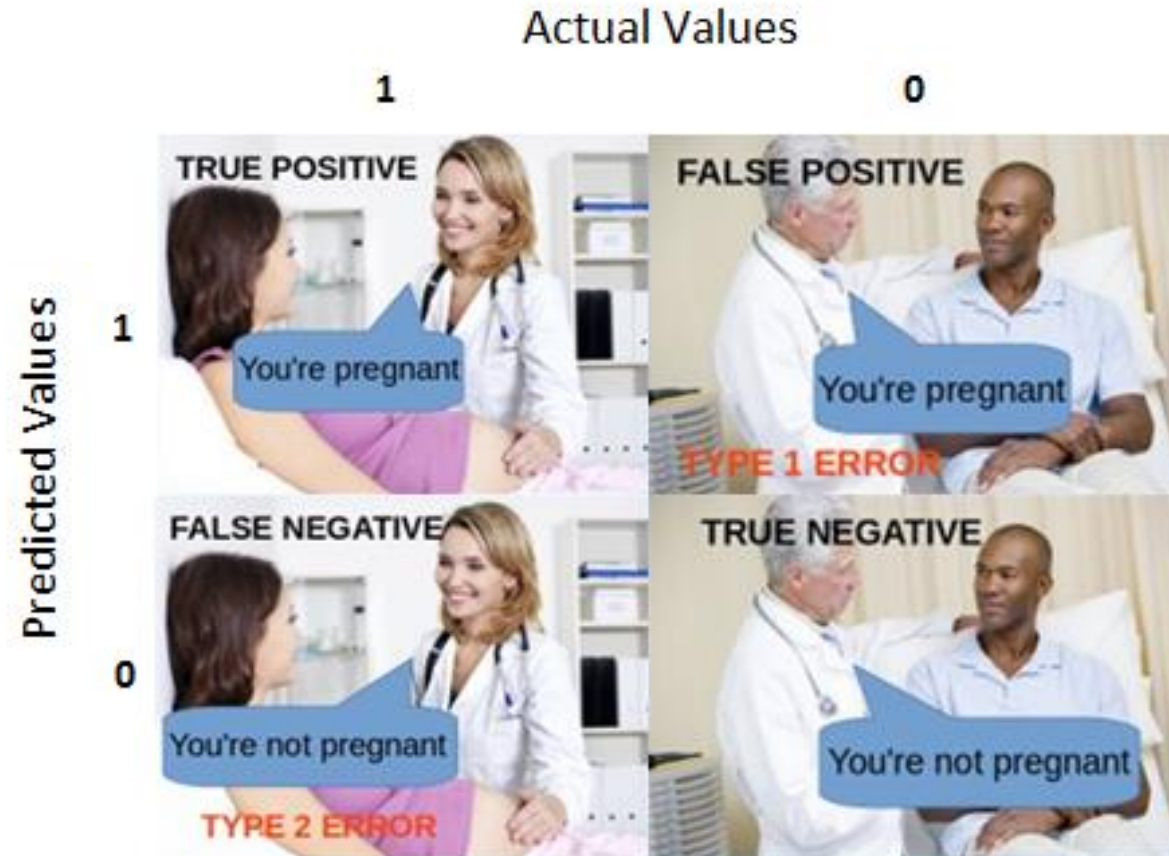
Confusion Matrix

		Predicted class	
		P	N
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)

We describe predicted values as Positive and Negative and actual values as True and False:



Confusion Matrix



Confusion Matrix

True Positive:

- Interpretation: You predicted positive and it's true. You predicted that a woman is pregnant and she actually is.

True Negative:

- Interpretation: You predicted negative and it's true. You predicted that a man is not pregnant and he actually is not.

False Positive: (Type 1 Error)

- Interpretation: You predicted positive and it's false. You predicted that a man is pregnant but he actually is not.

False Negative: (Type 2 Error)

- Interpretation: You predicted negative and it's false. You predicted that a woman is not pregnant but she actually is.

Confusion Matrix

		Predicted class	
		<i>P</i>	<i>N</i>
Actual Class	<i>P</i>	True Positives (TP)	False Negatives (FN)
	<i>N</i>	False Positives (FP)	True Negatives (TN)

The prediction accuracy can then be calculated directly from the confusion matrix:

$$Accuracy = \frac{TP + TN}{FP + FN + TP + TN}$$

In unbalance problem the accuracy is not descriptive.

Deducing performance metrics from a confusion matrix

Precision is the fraction of relevant instances among the retrieved instances

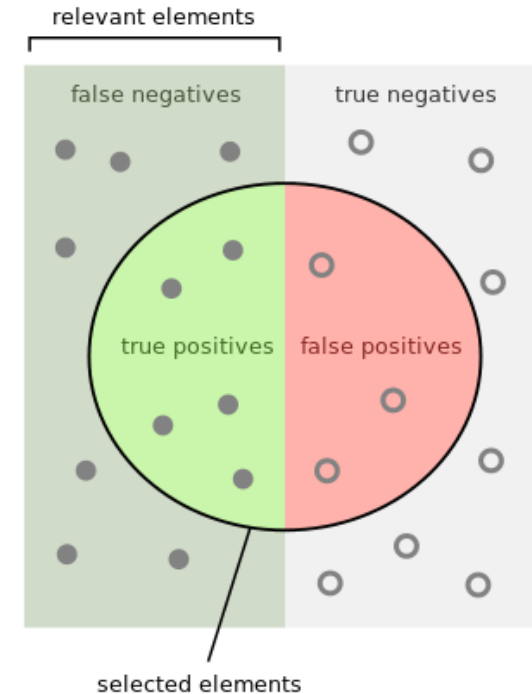
$$P = \frac{TP}{TP + FP}$$

Recall is the fraction of relevant instances that have been retrieved over the total amount of relevant instances.

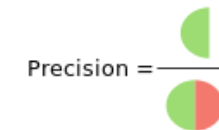
$$R = \frac{TP}{FN + TP}$$

F-Score is the *harmonic mean* of precision and recall

$$F - score = 2 \cdot \frac{P \cdot R}{P + R}$$



How many selected items are relevant?



How many relevant items are selected?

